

NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03

23 AUGUST 2018
2 hours

Candidates answer on the Question Paper.

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid. DO NOT WRITE IN ANY BARCODES.

Section A

Answer all questions in the spaces provided on the Question paper.

Section B

Answer one question in this section on writing papers provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [30]	
2 [10]	
3 [10]	
4 [25]	
5 [25]	
TOTAL P2 [35%]	75

This document consists of **14** printed pages and **0** blank pages.

[Turn over

Section A

Answer **all** questions in this section.

- 1 Foetal haemoglobin (HbF) is the main oxygen transport protein in foetuses during the last seven months of development in the uterus and persists in the newborn until six months old. Fig.1.1 shows a protein structure of HbF, which has a similar structure to HbA.

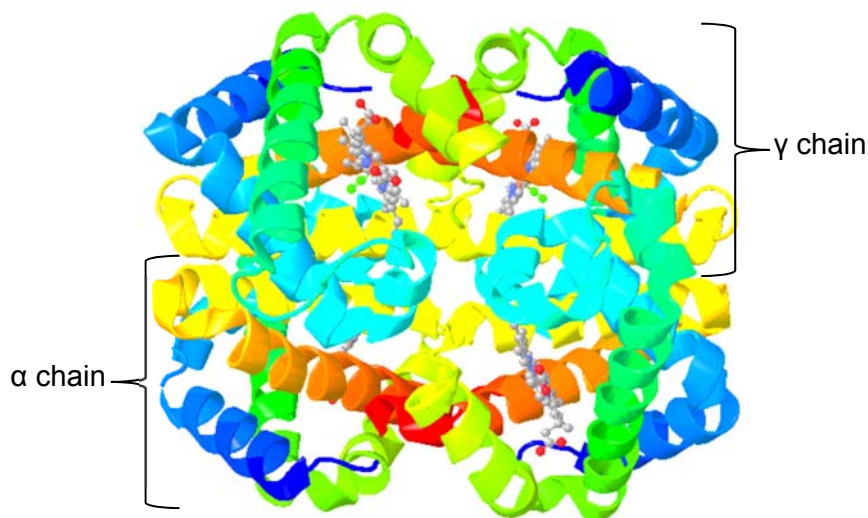


Fig. 1.1

- (a)(i) With reference to Fig.1.1, describe the features within HbF that are similar to HbA.

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- (a)(ii) HbF is a protein whose polypeptide chains are made up of amino acids that are joined together to achieve the final molecular configuration as shown in Fig. 1.1.

Using the monomer shown in Fig. 1.2, draw out a full labelled diagram to illustrate how the bond formation of bond is achieved between 2 such monomers.

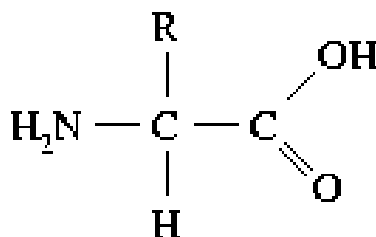


Fig. 1.2

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In newborns, foetal hemoglobin is completely replaced by adult hemoglobin (HbA) by approximately 6 months post-natal. Fig. 1.3 shows an oxygen saturation curve for HbF which appears left-shifted when compared to adult form (HbA). HbF is better able to bind to oxygen compared to HbA, giving better access to oxygen from the mother's bloodstream.

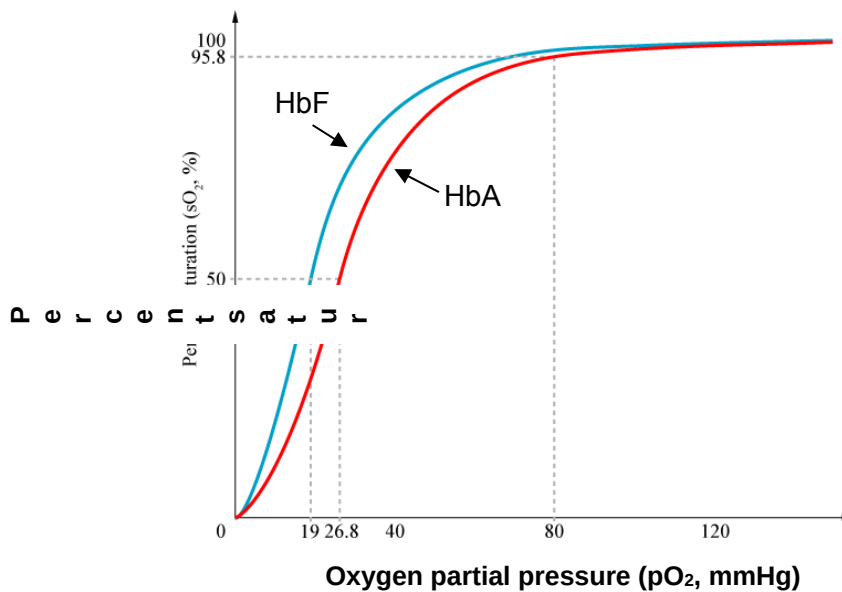


Fig.1.3

(a)(iii) With reference to Fig.1.3, suggest the significance of HbF possessing a higher capacity in binding to oxygen compared to HbA.

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Formation of globin chains varies in foetal to adult life. The alpha (α) and beta (β) and to a lesser extent the delta (δ) globin chains form HbA. In foetus, the gamma (γ), epsilon (ϵ), and zeta (ζ) chains are present. Fig. 1.4 shows the β -like globin genes that are expressed in a developmentally restricted manner, which is located on chromosome 11, as depicted in locus control region as well as the relative proportions of the various globin chains from conception to six months of age.

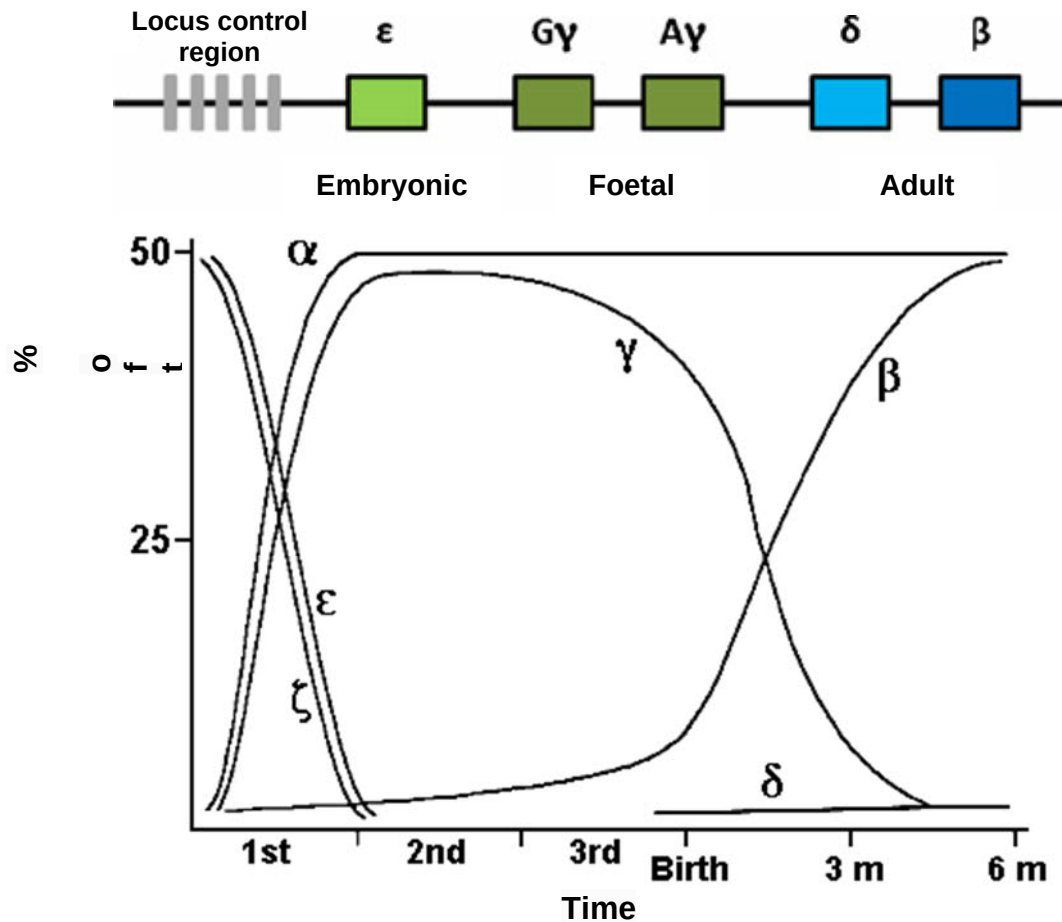


Fig. 1.4

(a)(iv) With reference to Fig.1.4, explain how the globin genes are regulated at transcriptional level to produce HbF and then HbA.

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Sickle cell anaemia is an inherited disease where the red blood cells are most commonly caused by the haemoglobin variant HbS, where a mutation has occurred at the beta globin polypeptide chain. Figure 1.5 shows the blast sequence of both the normal adult haemoglobin variant, HbA, as well as the mutant adult haemoglobin variant, HbS.

Beta globin sequence in normal adult haemoglobin (HbA)

nucleotide base	CTG ACT CCT GAG GAG AAG TCT
amino acid	Leu – Thr – Pro – Glu – Glu – Lys – Ser

Beta globin sequence in mutant adult haemoglobin (HbS)

nucleotide base	CTG ACT CCT GTG GAG AAG TCT
amino acid	Leu – Thr – Pro – Val – Glu – Lys – Ser

Fig. 1.5

(b)(i) With reference to Fig. 1.5, suggest how the mutation could have occurred.

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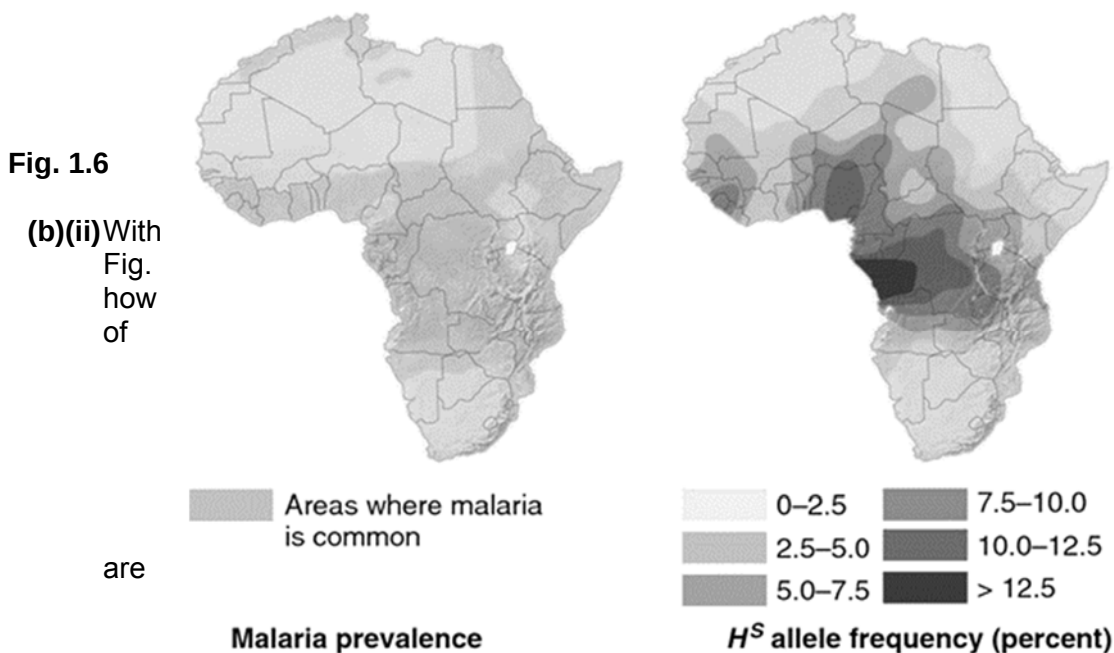
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Sickle cell disease has its highest occurrence in Sub-Saharan Africa. Despite the survival disadvantage of the allele HbS, HbS allele persists in the population at more than 12.5%.

Fig. 1.6 shows regions where sickle cell anaemia is most common, which coincide with regions where malaria is. Malaria-causing parasites are introduced into the blood by mosquitos.



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In adults, foetal hemoglobin production can be reactivated and induced pharmacologically, which is useful in the treatment of diseases such as sickle-cell disease. When present in individuals with sickle cell anaemia, HbF prevents sickling. Besides induced HbF treatment, haematopoietic stem cell transplantation (HSCT), usually derived from bone marrow, peripheral blood or umbilical cord blood has also shown great progress in the treatment of sickle cell disease recently. This involves in either the extraction of stem cells from various sources which may be autologous (i.e. using the patient's own stem cells are used), allogeneic (i.e. using stem cells derived from a donor) or syngeneic (i.e. using stem cells from an identical twin). Researchers cultivate these extracted stem cells which is then injected back into the patient. Such therapy for sickle cell anaemia could potentially give new hope to patients with sickle cell anaemia, although the treatment comes with its own risk.

(c)(i) Describe the unique features of stem cells used to treat this disease.

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(c)(ii) Explain why HSCT treatment from allogeneic sources may pose some risks to the patient.

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(c)(iii) Suggest why extraction of haematopoietic stem cells from autologous sources may not be as effective for therapy.

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(c)(iv) Umbilical cord blood has a higher concentration of haematopoietic stem cell than normally found in adult blood. Just in 2014, Singapore Cord Blood Bank (SCBB) received a 67% increase in cord blood donation compared to the year 2010. Nowadays, many expectant mothers and couples opt in banking their child's cord blood for many reasons.

Suggest the advantages of banking baby's cord blood.

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The signals governing self-renewal versus multi-lineage differentiation of hematopoietic stem cells (HSC) are complex. Several key signaling pathways including BMP, Hedgehog, Notch, Wnt, TGF-beta and others play important but different roles during foetal hematopoietic development, normal bone marrow haematopoiesis (i.e. formation of blood cellular components) and leukemogenesis (i.e. development of leukaemia). Haematopoiesis is controlled by combined effects of growth factors that permit cellular proliferation and transcription factors that activate lineage-specific genes. mRNA transcripts resulted from the these transcription of cell-specific genes must be successfully processed before they can be exported out of the nucleus, as shown in Fig. 1.7.

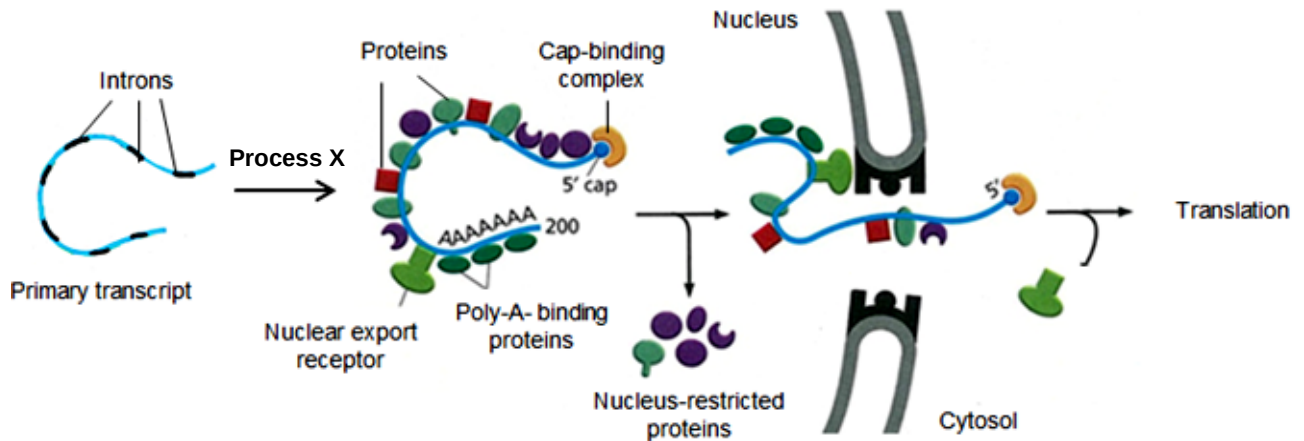


Fig. 1.7

With reference to Fig. 1.7,

(d)(i) Describe process X that results in the formation of a continuous mRNA coding sequence

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(d)(ii) Explain the significance of process X in regulating translation of mRNA.

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Translational control plays a crucial role in gene expression regulation, especially defining the proteome, maintaining homeostasis and controlling cell proliferation, growth and development. Translational control governs the efficiency of mRNAs. There are several ways in how mRNA translation can be regulated. One of the best studied examples include phosphorylation of eIF2 α (a subunit within eIF2 protein), eukaryotic initiation factor 2, which results in an inability for tRNA-binding methionine (met-tRNA) from binding to ribosomes. eIF2 is an essential factor for protein synthesis that forms an initial complex with initiator Met-tRNA, using GTP as energy source. This is shown in Fig. 1.8.

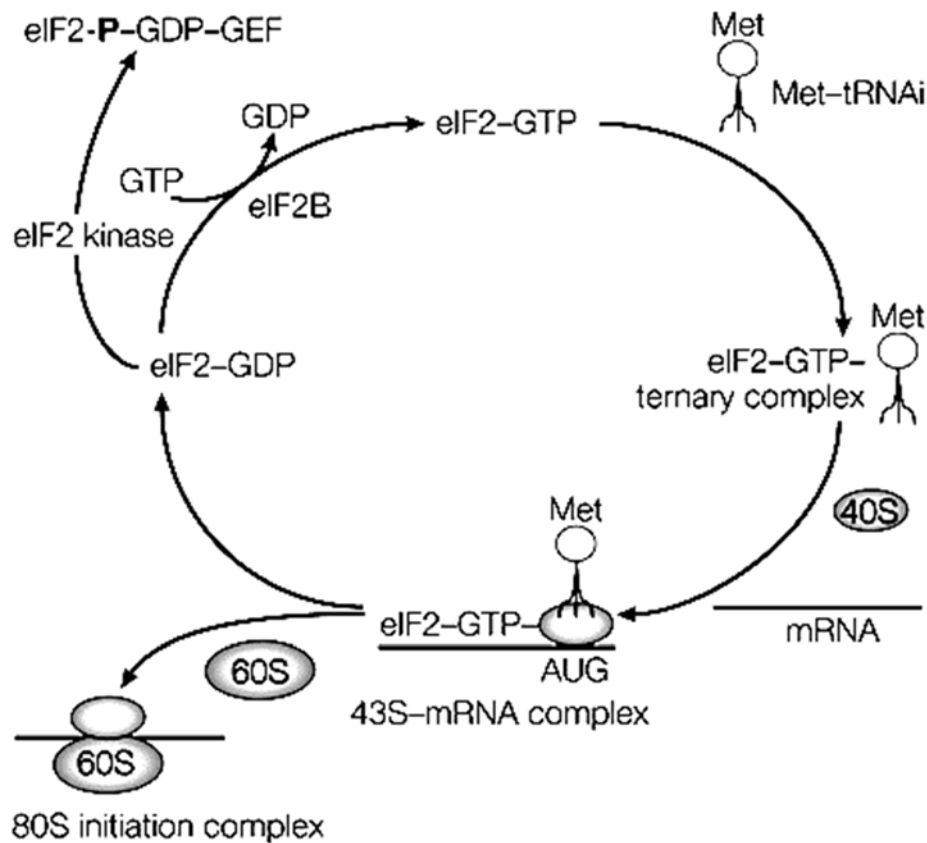


Fig. 1.8

(d)(iii) With reference to Fig. 1.8, suggest how phosphorylation of eIF2 α of eIF2 protein may affect initiation of translation.

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- (d)(iv) Suggest one other method that can be used to control translation and explain how it achieves the outcome.

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[Total: 30]

- 2 Huntington's disease is an inherited disease that causes the progressive breakdown (degeneration) of nerve cells in the brain. Huntington's disease has a broad impact on a person's functional abilities and usually results in movement, thinking (cognitive) and psychiatric disorders.

The data below shows the results of electrophoresis of PCR fragments amplified using primers for the site which has been shown to be altered in Huntington's disease. The inherited mutation in the Huntington's disease gene on chromosome 4, abnormally repeats the nucleotide sequence CAG from 36 to more than 120 times. The male parent, as shown by the black box, got Huntington's disease when he was 40 years old. Six of his children (individuals 3, 5, 7, 8, 10, 11) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band from the PCR fragment.

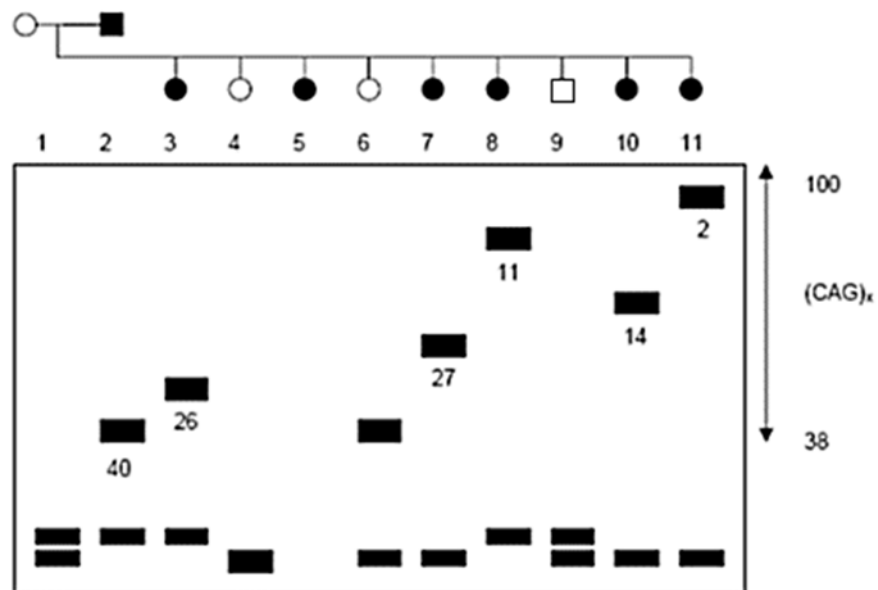


Fig. 2.1

- (a) With reference to Fig. 2.1, explain how Huntington's disease is inherited.

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(b) (i) What is the genotype of individual 4, explain why.

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(b)(ii) With reference to Fig. 2.1 describe a correlation between PCR fragment length and Huntington's disease. Explain your answer.

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(c) With reference to Fig. 2.1 explain the abnormality in individual 6.

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(d) Explain one further test on the product of Fig. 2.1 that can be employed to confirm the genotype of individual 6.

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[Total: 10]

- 3 Humans are hypothesized to share a common ancestor with modern great apes; however, humans have 23 pairs of chromosomes while great apes have 24 pairs. In 2005, one paper was published regarding the analysis of the genomes of humans and chimpanzees where they were compared for homology (Fig. 3.1). An evidence that strongly supports the shared ancestry is chromosome number 2 in humans, which was found to be the result of an end-to-end fusion of two ancestral chromosomes that remain separate in other great apes.

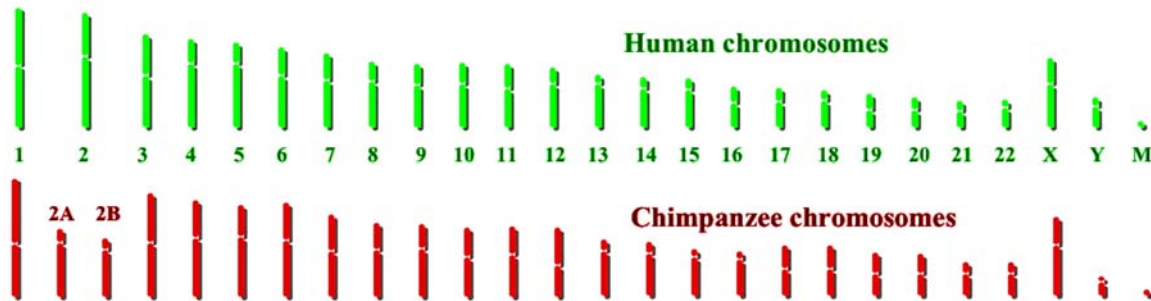


Fig. 3.1: Comparison of chromosomes 1 to 23 of humans and 1 to 24 of chimpanzees, M representing mitochondria.

- (a) Outline how fusion of chromosomes could possibly occur.

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- (b) Scientists hypothesised that humans and great apes are highly likely to share a common ancestry. By comparing the genome of humans against great apes, especially regarding the structure of chromosome number 2 in humans, they were able to find more evidences that the hypothesis should be true.

Predict what you would expect to find from this study if the hypothesis is true.

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- (c) Some animals such as mice and non-human primates (NHPs) are used in clinical research because of their relatedness to humans as seen in Fig. 3.2. However, there are strong concerns and strict regulations regarding the ethical use of NHPs in research, especially when compared to other animals. NHPs are used in clinical research only when deemed necessary, such as during the *Ebola* outbreak when they were used for further testing for vaccines after testing the same vaccines on mice.

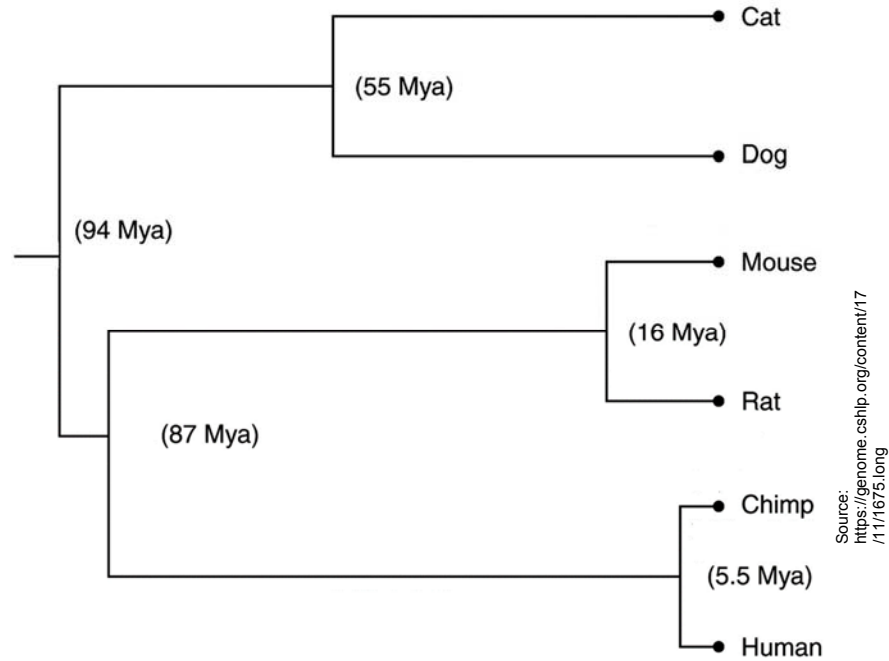


Fig. 3.2: Phylogenetic tree showing relative closeness of different mammals, where Mya = millions of years ago where lineage diverged.

- (i) Suggest why it is necessary to use non-human primates (NHPs) for further clinical research after testing on animals such as mice.

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- (ii) Discuss why the use of NHPs in scientific research raises such strong ethical concerns.

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[2]

[Total: 10]

Section B

Answer **one** question in this section

Write your answers on separate answer paper provided.
Answer each part on a **separate** piece of paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4** The actions of humans have had numerous impacts on both the environment and the natural world, affecting both plants and animals.

- (a)** One such impact is climate change which affects plants in many ways, including photosynthesis.

Outline the main stages of photosynthesis and suggest how anthropogenic climate change may potentially affect this process in plants. [14]

- (b)** Many modern domesticated animals are the result of artificial selection and breeding by humans over hundreds of years from wild animals. This includes chickens which were selected from jungle fowls for food and dogs which were selected from wolves to be kept as pets. Many of these domesticated animals look very different from their wild counterparts.

Explain how artificial selection has resulted in domesticated animals such as those mentioned above and comment on the view that “domesticated animals should be considered different species from their wild counterparts”. [11]

[Total: 25]

- 5** **(a)** Compare the advantages and disadvantages of the different reproductive cycles of lambda phage and influenza. [13]

- (b)** Suggest how phenotypic variation is brought about in viruses [12]

[Total: 25]

END OF PAPER